

10th Grade Learning Guide: Maximax and Maximin

Learning Criterion

C4: Applies the Minimax Regret criterion to make decisions without probabilities, selecting the alternative with the smallest possible maximum regret.

1 Minimax Regret

When probabilities are unavailable, the Minimax Regret criterion compares the regret associated with each alternative under each possible state of nature.

Focus	Maximum possible regret
First step	Find the best payoff in each state of nature
Second step	Construct the regret table
Third step	Find the maximum regret for each alternative
Final step	Choose the smallest maximum regret

First, identify the best payoff in each state of nature:

$$b(S_j) = \max_i P_{ay}(A_i, S_j)$$

Then calculate the regret for each alternative and state:

$$r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j)$$

For each alternative, find its maximum regret:

$$R(A_i) = \max_j r(A_i, S_j)$$

Finally, choose the alternative with the smallest maximum regret:

$$A_{\text{Minimax Regret}}^* = \arg \min_i R(A_i)$$

2 Solved Example: Clothing Store

A local clothing store is deciding whether to launch a new advertising campaign before the holiday season or continue

with its current marketing strategy. The manager knows that customer response to the campaign is uncertain and has no reliable information about the probability of strong or weak sales.

	S_1 : Strong Sales	S_2 : Weak Sales
Alternative	S_1	S_2
Launch new campaign (A_1)	55	5
Keep current strategy (A_2)	35	22

All payoffs are expressed in thousands of dollars. What decision should the manager make using the Minimax Regret criterion?

2.1 Solution

Step 1: Identify the alternatives and states of nature.

Extended opportunity loss table

Alternative	S_1	S_2	Maximum Regret
A_1 : Launch new campaign	0	17	$\max\{0, 17\} = 17$
A_2 : Keep current strategy	20	0	$\max\{20, 0\} = 20$

3 Guided Practice: Clothing Store

A clothing store must decide whether to launch a new advertising campaign or keep its current strategy. Sales could be strong or weak. All payoffs are in thousands of dollars.

Alternative	Strong sales	Weak sales
Launch new campaign	55	5
Keep current strategy	35	22

Step 1: Identify

$$\begin{aligned}
A_1 &= \underline{\hspace{2cm}} \\
A_2 &= \underline{\hspace{2cm}} \\
S_1 &= \underline{\hspace{2cm}} \\
S_2 &= \underline{\hspace{2cm}}
\end{aligned}$$

Step 2: Decide

Maximax

$$\begin{aligned}
M_1 &= \max\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}} \\
M_2 &= \max\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}} \\
A_{\text{Maximax}}^* &= \underline{\hspace{2cm}}
\end{aligned}$$

Maximin

$$\begin{aligned}
m_1 &= \min\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}} \\
m_2 &= \min\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}} \\
A_{\text{Maximin}}^* &= \underline{\hspace{2cm}}
\end{aligned}$$

Step 3: Complete the table

Alternative	Max	Min
A_1	<u> </u>	<u> </u>
A_2	<u> </u>	<u> </u>
A_{Maximax}^*	<u> </u>	<u> </u>
A_{Maximin}^*	<u> </u>	<u> </u>

4 Practice: Restaurant

A restaurant must decide whether to introduce a new lunch menu or keep its current menu. Demand could be high or low. All payoffs are in thousands of dollars.

Alternative	High demand	Low demand
New lunch menu	48	6
Current menu	30	21

Step 1: Identify

$$\begin{aligned}
A_1 &= \underline{\hspace{2cm}} & A_2 &= \underline{\hspace{2cm}} \\
S_1 &= \underline{\hspace{2cm}} & S_2 &= \underline{\hspace{2cm}}
\end{aligned}$$

Step 2: Decide

Maximax

$$\begin{aligned}
M_1 &= \max\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}} \\
M_2 &= \max\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}} \\
A_{\text{Maximax}}^* &= \underline{\hspace{2cm}}
\end{aligned}$$

Maximin

$$\begin{aligned}
m_1 &= \min\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}} \\
m_2 &= \min\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}} \\
A_{\text{Maximin}}^* &= \underline{\hspace{2cm}}
\end{aligned}$$

Step 3: Complete the table

Alternative	Max	Min
A_1	<u> </u>	<u> </u>
A_2	<u> </u>	<u> </u>
A_{Maximax}^*	<u> </u>	<u> </u>
A_{Maximin}^*	<u> </u>	<u> </u>

	A_1	A_2	A_{Maximax}^*	A_{Maximin}^*
Max	<u> </u>	<u> </u>	<u> </u>	<u> </u>
Min	<u> </u>	<u> </u>	<u> </u>	<u> </u>

5 Guided Practice: Technology Company

A technology company must choose a strategy for launching a new device. Market demand could be high, medium, or low. All payoffs are in thousands of dollars.

Alternative	Demand		
	High	Medium	Low
Premium version	80	35	-20
Standard version	60	42	18
Budget version	40	32	25
Delay launch	15	15	15

Step 1: Identify

A_1 = _____ A_2 = _____
 A_3 = _____ A_4 = _____
 S_1 = _____ S_2 = _____
 S_3 = _____

Alternative	Demand		
	High	Medium	Low
Large production	90	30	-15
Medium production	65	45	20
Small production	45	38	28
Maintain production	20	20	20

Step 2: Decide

Maximax

M_1 = max{____, ____} = ____
 M_2 = max{____, ____} = ____
 M_3 = max{____, ____} = ____
 M_4 = max{____, ____} = ____
 A_{Maximax}^* = _____

Maximin

m_1 = min{____, ____} = ____
 m_2 = min{____, ____} = ____
 m_3 = min{____, ____} = ____
 m_4 = min{____, ____} = ____
 A_{Maximin}^* = _____

Step 3: Complete the table

	A_1	A_2	A_3	A_4	Decision
Max	_____	_____	_____	_____	A_{Maximax}^* = _____
Min	_____	_____	_____	_____	A_{Maximin}^* = _____

6 Independent Practice: Sports Equipment Company

A sports equipment company must choose a production strategy for a new product. Demand could be high, medium, or low.

Step 1: Identify

A_1 = _____ A_2 = _____
 A_3 = _____ A_4 = _____
 S_1 = _____ S_2 = _____
 S_3 = _____

Step 2: Complete the analysis

Alternative	S_1	S_2	S_3	Max	Min
A_1	90	30	-15	_____	_____
A_2	65	45	20	_____	_____
A_3	45	38	28	_____	_____
A_4	20	20	20	_____	_____

Maximax

A_{Maximax}^* = _____

Maximin

A_{Maximin}^* = _____

7 Independent Practice: Bakery Company

A bakery company must choose a production strategy for a new product. Demand could be high, medium, or low.

Alternative	Demand		
	High	Medium	Low
Large production	85	32	-10
Medium production	62	48	18
Small production	44	36	30
Maintain production	22	22	22

Step 1: Identify

A_1 = _____ A_2 = _____
 A_3 = _____ A_4 = _____
 S_1 = _____ S_2 = _____
 S_3 = _____

Step 2: Complete the analysis

Alternative	S_1	S_2	S_3	Max	Min
A_1	85	32	-10	_____	_____
A_2	62	48	18	_____	_____
A_3	44	36	30	_____	_____
A_4	22	22	22	_____	_____

Maximax

$$A_{\text{Maximax}}^* = \underline{\hspace{2cm}}$$

Maximin

$$A_{\text{Maximin}}^* = \underline{\hspace{2cm}}$$

8 Independent Practice: Furniture Company

A furniture company must choose a production strategy for a new collection. Demand could be high, medium, or

low.

Alternative	Demand		
	High	Medium	Low
Large production	95	28	-18
Medium production	70	50	16
Small production	48	40	32
Maintain production	25	25	25

Step 1: Identify

A_1 = _____ A_2 = _____
 A_3 = _____ A_4 = _____
 S_1 = _____ S_2 = _____
 S_3 = _____

Step 2: Complete the analysis

Alternative	S_1	S_2	S_3	Max	Min
A_1	95	28	-18	_____	_____
A_2	70	50	16	_____	_____
A_3	48	40	32	_____	_____
A_4	25	25	25	_____	_____

Maximax

$$A_{\text{Maximax}}^* = \underline{\hspace{2cm}}$$

Maximin

$$A_{\text{Maximin}}^* = \underline{\hspace{2cm}}$$